

Unified SunSpec Modbus Map for AE Inverters

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Introduction

Advanced Energy's line of solar power inverters present certain measurement and control information via their Modbus interfaces. Each product line (NX, TX, 3TL) has its own Modbus map with different address ranges and different information. Even within a product line, there are variations based on model number (e.g., 500NX vs. 1000NX) or firmware versions within a model (e.g., 500NX Rev J firmware and later, vs. pre Rev J firmware). Also, the word order conventions within the Modbus register space are different between NX and TX products (NX is little-endian and TX is big-endian).

As part of product line consolidation and unification, having a common Modbus map across the entire product line would be advantageous for customers, Field Service and Engineering. Creating a common map could be done in several ways: (1) pick one of the existing product maps and implement that on the other product lines, (2) create a new custom map using parts of the existing maps or (3) use a common industry standard map.

The SunSpec Alliance (<http://www.sunspec.org>) has developed a set of standard Modbus maps (called "models") with consensus from inverter manufacturers, end users and other interested parties. These maps include support for basic inverter measurements, extended measurements, nameplate ratings, set points and limits, as well as advanced inverter controls. The SunSpec models are design to be used in combination with each other and for discovery by a SCADA controller or PMU/EMS so the controlling entity can determine which models are supported by a given product. In addition, within a given model, there are both mandatory and optional fields so not all registers have to be populated for cases where a particular measurement or control is not supported by the device.

It is my recommendation that AE implement support for the standardized SunSpec models across the product line as time and resources permit. The following sections detail the minimal and extended set of SunSpec models that should be included in our inverters.

References

- [1] SunSpec-Alliance-Specification-Common-Elements-v1.5-DRAFT-9, November 1, 2012
- [2] SunSpec Alliance - Inverter Controls DRAFT 13, January 10, 2014
- [3] SunSpec Alliance PICS, August 13, 2013

SunSpec Mapping Overview

The SunSpec Alliance Interoperability Specifications describe the information models and MODBUS register mappings for devices used in Renewable Energy systems. A physical device, such as an inverter, is represented by a collection of one or more of these logical models. This document describes the Common Models and model design conventions. All SunSpec devices must contain the device information common model that describes this particular device as the first logical block of the device.

All SunSpec devices support a consecutive collection of well-defined blocks, each prefaced with a well-known Device Identity and length field. This allows a remote client to browse the contents skipping blocks with unrecognized Device Identity (ID) values. The Common Model specifically allows for extensibility in the following ways:

1. Definition of new Device Specific Blocks. New devices are defined and assigned a new device model id.
2. Revisions of Device Specific Blocks. Revisions to device models are defined and assigned a new device model id. Simple models may be appended to without affecting backwards compatibility and do not require a new model id. Readers shall rely on the length of the model to determine this.
3. Vendor specific extensions. Models may include fields that have vendor specific values. An example of this is the status and event code fields. Vendors may also define a Vendor Extension Block that includes fields and values specific to the vendor. These are assigned a device type id. This block may be then concatenated to the associated Device Specific Block.
4. Composite device types. Device Specific Blocks may be concatenated together to form a composite device. An example of this is a Weather Station represented by a collection of environmental models.
5. Device aggregations. A map may contain multiple Common Blocks. Each Common Block marks the start of a new device. This is useful when a map represents an aggregation of devices. (See the Module/Panel specification for an example of this usage.)

All SunSpec devices must include at least three blocks:

- The first block is the Common Block, which supplies vendor and model information for the device.
- The second (and subsequent) blocks will be device specific – for example an inverter block, or a inverterblock followed by one or more inverter control blocks.
- The final End Block formally marks the end of the Device Specific Blocks.

Example SunSpec Device with two device-specific blocks

Modbus Address	Name	Sunspec Model ID
4x4000	Common Model	
4x4007	Inverter Model	103
4x4012	Nameplate Ratings	120
4x4014	Basic Settings	
4x4017	Extended Meas.	122
4x4020	Immediate Controls	123
4x4022	Other Control Model	
4x40xxx	Other Control Model	
4x40xxx	End Model	

The SunSpec specifications also specify the allowed data types, the word order for multi-word values and the “not implemented” value for each data type.

Guide to AE Supported Models

In the tables below, each model is tagged with the Required or Optional tag. Required models should be supported by each inverter implementing the AE SunSpec compatible map. Optional models will be included or excluded based on inverter capability. Even when a model is included, not all fields in the model are mandatory. Refer to the SunSpec PICS document or source XML files for the detailed description of each of these models, including labels, units and mandatory/optional tags.

In the tables below, the inverter family columns indicate if support is currently available for the given element.

- X** in the column indicates it is supported (for NX and TX, this applies only to inverters using a PVM2021 or PVM2025 data monitoring board);
- U** indicates unimplemented and not available.

Common Model (1)

The Common Block model (1) would be common to all products. The Common model must start at one of three well-known addresses to allow for discovery by the controller. The starting address can be 0, 40000 or 50000. All three AE inverter product families have legacy Modbus maps that start at 0, so that is not a good selection. The AE1000NX currently implements a partial SunSpec compatible map starting at address 40000. This map consists of the Common Model, the AE DCS Model and an End block. This address range could be used, with other models added after the DCS model. This is the recommended option. (Note in inverters without a DCS, the DCS model would not be present and the first model after the common block would be the 3-Phase inverter model. The order of the models in the map is not prescribed.) The third option of using a start address of 50000 presents the problem with the AE1000NX of having two SunSpec Modbus maps, which could confuse a controller.

Register No.	Parameter	Data Type	R/W	Description and Measurement
SunSpec Common Model				
0, 1	SunSpec_ID	uint32	R	Unique SunSpec identifier 0x53756e53 (SunS)
2	ID	uint16	R	Common model id (1)
3	L	uint16	R	Length of common model (66)
4 - 19	Mn	String(32)	R	Manufacturer name (AEI)
20 - 35	Md	String(32)	R	Manufacturer model
36 - 43	Opt	String(16)	R	Manufacturer options
44 - 51	Vr	String(16)	R	Manufacturer version
52 - 67	SN	String(32)	R	Manufacturer serial number
68	DA	uint16	R	Device address (not used, reads as 65535)
69	Pad	pad16	R	Pad word (reads as 65535)

The manufacturer model name (Md), options (Opt), version (Vr) and serial number (SN) will vary from product to product.

Basic 3-Phase Inverter Model (103)

The basic 3-phase inverter model (103) would be common to all products.

Reg. No.	Parameter	Data Type	R/W	Description and Measurement	3TL	NX	TX
SunSpec 3-Phase Inverter Model							
0	ID	uint16	R	Inverter model id (103)	103	X	X
1	L	uint16	R	Length of inverter model (50)	50	X	X
2	A	uint16	R	Amps Sum of all phases	P1124.0*10	X	X
3	AphA	uint16)	R	Amps phase A RMS	P1124.1*10	X	X
4	AphB	uint16	R	Amps phase B	P1124.2*10	X	X
5	AphC	uint16	R	Amps phase C	P1124.3*10	X	X
6	A_SF	sunsf	R	Amps scale factor	-1	X	X
7	PPVphAB	uint16	R	Phase voltage AB	P1130.0*10	X	X
8	PPVphBC	uint16	R	Phase voltage BC	P1130.1*10	X	X
9	PPVphCA	uint16	R	Phase voltage CA	P1130.2*10	X	X
10	PhVphA	uint16	R	Phase voltage AN	P1123.1*10	U	U
11	PhVphB	uint16	R	Phase voltage BN	P1123.2*10	U	U
12	PhVphC	uint16	R	Phase voltage CN	P1123.3*10	U	U
13	V_SF	sunsf	R	Voltage scale factor	-1	X	X
14	W	uint16	R	Watts	P1106.0/10	X	X
15	W_SF	sunsf	R	Watts scale factor	1	X	X
16	Hz	uint16	R	Hertz Average all three phases	P115?.4*100	X	X
17	Hz_SF	sunsf	R	Hertz scale factor	-2	X	X
18	VA	uint16	R	Volt-Amps	P1106.2/10	U	U
19	VA_SF	sunsf	R	Volt-Amps scale factor	1	U	U
20	VAR	int16	R	Volt-Amp Reactive (var)	P1106.1/10		
21	VAR_SF	sunsf	R	Volt-Amp Reactive scale factor	X	U	
22	PF	int16	R	Power factor	1	X	X
23	PF_SF	sunsf	R	Power factor scale factor	X	X	X
24, 25	WH	acc32	R	Watt-Hours	X	X	X
26	WH_SF	sunsf	R	Watt-Hours scale factor	P1151	X	X
27	DCA	uint16	R	DC Amps	-2	X	X

Reg. No.	Parameter	Data Type	R/W	Description and Measurement	3TL	NX	TX
28	DCA_SF	sunsf	R	DC Amps scale factor	P1105.0*10	X	X
29	DCV	uint16	R	DC Volts	-1	X	X
30	DCV_SF	sunsf	R	DC Volts scale factor	P1104.0*10	X	X
31	DCW	uint16	R	DC Watts	-1	X	X
32	DCW_SF	sunsf	R	DC Watts scale factor	P1107.0/10	X	X
33	TmpCab	int16	R	Cabinet temperature	1	X	X
34	TmpSnk	int16	R	Heat Sink temperature	P92.1 (time)	X	X
35	TmpTrns	int16	R	Transformer temperature	P92.0 (time)	U	X
36	TmpOt	int16	R	Other temperature	NA	U	X
37	Tmp_SF	sunsf	R	Temperature scale factor	NA	X	X
38	St	enum16	R	Operating state	-1	X	X
39	StVnd	enum16	R	Vendor operating state	P501.0	X	X
40, 41	Evt1	bitfield32	R	Event1	X	X	X
42, 43	Evt2	bitfield32	R	Event2	X	X	X
44, 45	EvtVnd1	bitfield32	R	Vendor event bitfield 1	X	X	X
46, 47	EvtVnd2	bitfield32	R	Vendor event bitfield 2	X	X	X
48, 49	EvtVnd3	bitfield32	R	Vendor event bitfield 3	X	X	X
50, 51	EvtVnd4	bitfield32	R	Vendor event bitfield 4	X	X	X

Inverter Controls Nameplate Model (120)

The inverter nameplate model (120) would be common to all products. Note that the parameter names are typically taken directly from IEC 61850-90-7, which in turn came from the IEC 61850-7-4 and other IEC standards.

Reg. No.	Parameter	Data Type	R/W	Description and Measurement	3TL	NX	TX
Inverter Controls Nameplate Model							
0	ID	uint16	R	SunSpec IC nameplate model id (120)	X	X	X
1	L	uint16	R	Length of model (26)	X	X	X
2	DERType	enum16	R	Type of DER device. Default value is 4 to indicate PV device.	4	X	X
3	WRtg	uint16	R	Continuous power output capability of the inverter.	P1450.0	X	X

Reg. No.	Parameter	Data Type	R/W	Description and Measurement	3TL	NX	TX
4	WRtg_SF	sunssf	R	Scale factor	-1	X	X
5	VARtg	uint16	R	Continuous Volt-Ampere capability of the inverter.	P1450.0	X	U
6	VARtg_SF	sunssf	R	Scale factor	-1	X	U
7	VARRtgQ1	int16	R	Continuous VAR capability of the inverter in quadrant 1.	P1021.0*10	X	U
8	VARRtgQ2	int16	R	Continuous VAR capability of the inverter in quadrant 2.	P1021.0*10	U	U
9	VARRtgQ3	int16	R	Continuous VAR capability of the inverter in quadrant 3.	0	U	U
10	VARRtgQ4	int16	R	Continuous VAR capability of the inverter in quadrant 4.	0	X	U
11	VARRtg_SF	sunssf	R	Scale factor	-1	X	X
12	ARtg	uint16	R	Maximum RMS AC current level capability of the inverter.	P982.0*10	X	U
13	ARtg_SF	sunssf	R	Scale factor		X	X
14	PFRtgQ1	int16	R	Minimum power factor capability of the inverter in quadrant 1.	-1	X	X
15	PFRtgQ2	int16	R	Minimum power factor capability of the inverter in quadrant 2.	-	U	U
16	PFRtgQ3	int16	R	Minimum power factor capability of the inverter in quadrant 3.	-	U	U
17	PFRtgQ4	int16	R	Minimum power factor capability of the inverter in quadrant 4.	-	X	X
18	PFRtg_SF	sunssf	R	Scale factor	-	X	X
19	WHRtg	uint16	R	Nominal energy rating of storage device.	X	U	U
20	WHRtg_SF	sunssf	R	Scale factor	U	U	U
21	AhrRtg	uint16	R	The useable capacity of the battery. Maximum charge minus minimum charge from a technology capability perspective (Amp-hour capacity rating).	U	U	U

Reg. No.	Parameter	Data Type	R/W	Description and Measurement	3TL	NX	TX
22	AhrRtg_SF	sunssf	R	Scale factor for amp-hour rating.	U	U	U
23	MaxChaRte	uint16	R	Maximum rate of energy transfer into the storage device.	U	U	U
24	MaxChaRte_SF	sunssf	R	Scale factor	U	U	U
25	MaxDisChaRte	uint16	R	Maximum rate of energy transfer out of the storage device.	U	U	U
26	MaxDisChaRte_SF	sunssf	R	Scale factor	U	U	U
27	Pad	pad	R	Pad register.	U	X	X

Inverter Controls Extended Measurements Model (122)

The inverter extended measurements and status model (122) would be common to all products.

Register No.	Parameter	Data Type	R/W	Description and Measurement	3TL	NX	TX
Inverter Controls Extended Measurements and Status Model							
0	ID	uint16	R	SunSpec IC measurements model id (122)	X	X	X
1	L	uint16	R	Length of model (44)	X	X	X
2	PVConn	bitfield16	R	PV inverter present/available status. Enumerated value.	X	X	X
3	StorConn	bitfield16	R	Storage inverter present/available status. Enumerated value.	X	X	X
4	ECPCConn	bitfield16	R	ECP connection status: disconnected=0 connected=1.	X	X	X
5-8	ActWh	acc64	R	AC lifetime active (real) energy output.	X	X	X
9-12	ActVAh	acc64	R	AC lifetime apparent energy output.	U	U	U
13-16	ActVARhQ1	acc64	R	AC lifetime reactive energy output in quadrant 1.	U	U	U
17-20	ActVARhQ2	acc64	R	AC lifetime reactive energy output in quadrant 2.	U	U	U

Register No.	Parameter	Data Type	R/W	Description and Measurement	3TL	NX	TX
21-24	ActVArhQ3	acc64	R	AC lifetime negative energy output in quadrant 3.	U	U	U
25-28	ActVArhQ4	acc64	R	AC lifetime reactive energy output in quadrant 4.	U	U	U
29	VArAval	int16	R	Amount of VARs available without impacting watts output.	U	U	U
30	VArAval_SF	sunssf	R	Scale factor for available VARs.	U	U	U
31	WAval	uint16	R	Amount of Watts available.	U	U	U
32	WAval_SF	sunssf	R	Scale factor for available Watts.	U	U	U
33,34	StSetLimMask	bitfield32	R	Bit Mask indicating setpoint limit(s) reached.	U	U	U
35,36	StActCtl	bitfield32	R	Bit Mask indicating which inverter controls are active.	U	U	U
37-40	TmSrc	string	R	Source of time synchronization.	X	X	X
41,42	Tms	uint32	R	Seconds since 01-01-2000 00:00 UTC	X	X	X
43	RtSt	bitfield16	R	Bit Mask indicating active ride-through status.	U	U	U
44	Ris	uint16	R	Isolation resistance.	X	U	U
45	Ris_SF	sunssf	R	Scale factor for isolation resistance.	X	U	U

Inverter Immediate Controls Model (123)

The inverter immediate controls model (123) would be common to all products.

Not yet implemented for 3TL products

Reg. No.	Parameter	Data Type	R/W	Description and Measurement	3TL	NX	TX
Inverter Controls Immediate Controls Model							
0	ID	uint16	R	SunSpec IC nameplate model id (123)	X	X	X
1	L	uint16	R	Length of model (24)	X	X	X

Reg. No.	Parameter	Data Type	R/W	Description and Measurement	3TL	NX	TX
2	Conn_WinTms	uint16	RW	Time window for connect/disconnect.	U	U	U
3	Conn_RvrtTms	uint16	RW	Timeout period for connect/disconnect.	U	U	U
4	Conn	enum16	RW	Enumerated valued. Connection control.	X	X	X
5	WMaxLimPct	uint16	RW	Set power output to specified level.	X	X	X
6	WMaxLimPct_WinTms	uint16	RW	Time window for power limit change.	U	U	U
7	WMaxLimPct_RvrtTms	uint16	RW	Timeout period for power limit.	U	U	U
8	WMaxLimPct_RmpTms	uint16	RW	Ramp time for moving from current setpoint to new setpoint.	X	X	X
9	WMaxLim_Ena	enum16	RW	Enumerated valued. Throttle enable/disable control.	X	X	X
10	OutPFSet	int16	RW	Set power factor to specific value - cosine of angle.	X	X	X
11	OutPFSet_WinTms	uint16	RW	Time window for power factor change.	U	U	U
12	OutPFSet_RvrtTms	uint16	RW	Timeout period for power factor.	U	U	U
13	OutPFSet_RmpTms	uint16	RW	Ramp time for moving from current setpoint to new setpt.	X	X	X
14	OutPFSet_Ena	enum16	RW	Enumerated valued. Fixed power factor enable/disable control.	X	X	X
15	VArWMaxPct	int16	RW	Reactive power in percent of WMax.	U	U	U
16	VArMaxPct	int16	RW	Reactive power in percent of VArMax.	X	X	U
17	VArAvalPct	int16	RW	Reactive power in percent of VArAval.	U	U	U

Reg. No.	Parameter	Data Type	R/W	Description and Measurement	3TL	NX	TX
18	VArPct_WinTms	uint16	RW	Time window for VAR limit change.	U	U	U
19	VArPct_RvrtTms	uint16	RW	Timeout period for VAR limit.	U	U	U
20	VArPct_RmpTms	uint16	RW	Ramp time to move from current setpoint to new setpt.	X	X	U
21	VArPct_Mod	enum16	RW	Enumerated value. VAR percent limit mode.	X	X	X
22	VArPct_Ena	enum16	RW	Enumerated valued. Percent limit VAR enable/disable control.	X	X	X
23	WMaxLimPct_SF	sunssf	R	Scale factor for power output percent.	X	X	X
24	OutPFSet_SF	sunssf	R	Scale factor for power factor.	X	X	X
25	VArPct_SF	sunssf	R	Scale factor for reactive power percent.	X	X	X

AE Specific Models

Advanced Energy has been assigned a block of vendor specific model ids for use with our vendor specific maps. Currently, the only vendor specific map that has been developed is the one for the DC Subsystem used with the AE1000NX.

Specific Model NX / TX (64070)

Reg.No.	Parameter	Data Type	R/W	Description and Measurement	3TL	NX	TX
AE DC Subsystem Model							
0	ID	uint16	R	AE DCS model id (64070)	U	X	U
1	L	uint16	R	Length of model (66)	U	X	U
Fixed Block							
2	DCA_SF	sunssf	R	Scale factor for current.	U	X	U
3	DCV_SF	sunssf	R	Scale factor for voltage.	U	X	U
4	N	uint16	R	Number of contactors (10).	U	X	U
5	StCtcr	bitfield16	R	Contactor status.	U	X	U
6	StCpt	enum16	R	CPT connection status.	U	X	U
7,8	EvtFlt	bitfield32	R	DCS Faults. Bitfield.	U	X	U
9,10	EvtWrn	bitfield32	R	DCS Warnings. Bitfield.	U	X	U
11	Rsvd	uint16	R	Reserved.	U	X	U
12	State	enum16	R	DCS controller state.	U	X	U
13	DCA	int16	R	Aggregate DC current, all arrays	U	X	U
14	DCADif	int16	R	Aggregate differential current.	U	X	U
15	DCV	int16	R	Aggregate DC voltage, all arrays.	U	X	U
16	Tmp	int16	R	DCS board temperature	U	X	U
Repeating Block (10 instances)							
17	InID	uint16	R	ID number for this array.	U	X	U
18	InFlt	bitfield16	R	Fault word for this array.	U	X	U
19	InWrn	bitfield16	R	Warning word for this array.	U	X	U
29	InPDCA	int16	R	Positive DC current for array.	U	X	U
21	InNDCA	int16	R	Negative DC current for array.	U	X	U
Fixed Block							

67	Pad	pad	R	Pad word	U	X	U
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Advanced Energy Specific Model 3TL (64107)

Reg. No.	Parameter	DataType	R/W	Description	3TL
0	ID	uint16	R	Sunspec model ID (64107)	
1	L	uint16	R	Length of model (60 ?)	
2	Temp C Right	sint16	R	Temperature Cooler Right	92.0
3	Temp C Left	sint16	R	Temperature Cooler Left	92.3
4	Temp i Right	sint16	R	Temperature interior lower right	92.2
5	Temp i Left	sint16	R	Temperature interior upper left	92.1
6	Temp scale factor	sunfs	R	Temperatur scale factor (-1)	
7	Iso Meas Rest NZK	u32	R	Iso Measurement Measured Resistance (Negative ZK)	112.0
9	Iso Meas Rest PZK	u32	R	Iso Measurement Mesured Resistance (Positive ZK)	112.1
11	Iso Meas scale	sunfs	R	Iso Measurement scale factor (1)	
12	Reactive PW SP	float	R	Reactive Power SetPoint	1020.0
14	Reactive PW MX	float	R	Max Reactive Power	1021.0
16	Desired RPW	sint32	R/W	Desired Reactive Power	1159.0
18	Des RPW scale	sunfs	R	Desired Reactive Power scale factor (-1)	
19	Desired Cos Phi	Uint16	R/W	Desired Reactive Power Cos Phi	1167.0
20	DesCosPhiScale	sunfs	R	Desired Reactive Power scale factor (-2)	
21	Active PW Out	float	R	MPPT Active Power output	1065.0
23	Remote Max PW	uint16	R/W	Remote Max Power	1162.0
24	RemMaxPW scale	sunfs	R	Remote Max Power scale factor (-1)	
25	Irradiation energy	u32	R	Daily sun irradiation energy in kWh/m ²	1156.0
27	Irradition factor	sunfs	R	Temperatur scale factor (-4)	

Reg. No.	Parameter	DataType	R/W	Description	3TL
0	ID	uint16	R	Sunspec model ID (64107)	
1	L	uint16	R	Length of model (60 ?)	
2	Temp C Right	sint16	R	Temperature Cooler Right	92.0
3	Temp C Left	sint16	R	Temperature Cooler Left	92.3
4	Temp i Right	sint16	R	Temperature interior lower right	92.2
5	Temp i Left	sint16	R	Temperature interior upper left	92.1
6	Temp scale factor	sunfs	R	Temperatur scale factor (-1)	
7	Iso Meas Rest NZK	u32	R	Iso Measurement Measured Resistance (Negative ZK)	112.0
28	TransitionStTime	uint16	R	Remaining Time to connect in seconds	1500.0
29	TransitionSTDC	uint16	R	Transition Status for DC: 0 - Ok 1 - Ok Waiting (Time not elapsed) 2 - Under Voltage 3 - Over Voltage	1500.1
30	TransitionSTAC	uint16	R	Transition Status for AC: 0 - Ok 1 - Ok Waiting (Time not elapsed) 2 - Under Voltage 3 - Over Voltage 4 - Under Frequency 5 - Over Frequency 6 - Low Power	1500.2
31	UserClr	uint32	R	UserClearance 0 = inverter stays in check state 1 = normal activation (Writeable via USS)	990.0
33	PortalRqCmd	uint32	R/W	Portal Request Command. Set to 1 to enforce a Portal update via Ethernet.	477.0
35	ClrPerstFail	uint32	R/W	Set to 424242 to remove any blocking errors.	2002.0
37	LatestErrId	uint32	R	The latest error id code of an error occurred and present.	500.0

Reg. No.	Parameter	DataType	R/W	Description	3TL
0	ID	uint16	R	Sunspec model ID (64107)	
1	L	uint16	R	Length of model (60 ?)	
2	Temp C Right	sint16	R	Temperature Cooler Right	92.0
3	Temp C Left	sint16	R	Temperature Cooler Left	92.3
4	Temp i Right	sint16	R	Temperature interior lower right	92.2
5	Temp i Left	sint16	R	Temperature interior upper left	92.1
6	Temp scale factor	sunfs	R	Temperatur scale factor (-1)	
7	Iso Meas Rest NZK	u32	R	Iso Measurement Measured Resistance (Negative ZK)	112.0
39	ArrWrn	Bitfield32	R	Array warnings	
41	GridWrn	Bitfield32	R	Grid warnings	
43	SysWrn	Bitfield32	R	System warnings	
45	DrvWarn	Bitfield32	R	Drive warnings	
47	TempWarn	Bitfield32	R	Temperature warnings	
49	SysFault	Bitfield32	R	System faults	
51	NRFault	Bitfield32	R	Non Restartable Faults	
53	OpTime	uint32	R	Operation Time in s	1152.0
55	ExtSens1	sint16	R	Value of External Sensor 1	1191.0
56	ExtSens2	sint16	R	Value of External Sensor 1	1193.0
57	DailyEnergy	Uint32	R	Energy production for current day in Wh * 100	1150.0
59	DailyEnergySf	sunsf	R	2	
60	StartUpdate	uint16	W	Write anything to start the update, after the update-file was uploaded	
61	UpdProgr	uint16	R	Update Progress. 1-100 for progress in % Or Negative number for update error code.	

Other SunSpec Inverter Controls Models (Optional)

Many vendors will also be implementing the Basic Settings profile (121), Ride-through profiles: LVRT (129), HVRT (130), LFRT (135), HFRT (136) and Volt-VAr profile (126) in their Modbus maps. In the case of AE inverters with the PVM2021 or PVM2025, the relevant content from these profiles is available via the PVM's web user interface, so at least initially, there is no plan to include them in the Modbus map.

In addition to the above models which are currently (or soon will be) supported by one or more AE products, the following models have been defined for future use: Basic Storage Controls (124), Pricing Signal (125), Parameterized Frequency-Watt (127), Dynamic Reactive Current (128), Watt-PF (131), Volt-Watt (132), Basic Scheduling (133), Curve-Based Frequency Watt (134), LVRT Must Remain Connected (137), HVRT Must Remain Connected (138).

SunSpec has also defined a set of Network Configuration models, which allows a controller to read or set information about the various network interfaces on a device. These should be considered for inclusion in any SunSpec compatible product.